

Sipeng Zheng

Partner, Research Scientist | Embodied AI and Multimodal Foundation Models

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I am a Partner and Research Scientist at BeingBeyond, where I lead the Embodied Multimodal Pretraining team and work closely with Prof. Zongqing Lu on foundation models for embodied AI. I lead the Being-H series, including Being-H0, Being-H0.5, and Being-H0.7, and my recent work also spans Being-M and Being-VL, with emphasis on vision-language-action pretraining, human motion generation, visual tokenization, and robot control. Previously, I was a Research Scientist at the Beijing Academy of Artificial Intelligence (BAAI). I received my Ph.D. and B.S. in Computer Science from Renmin University of China in 2023 and 2018, respectively, advised by Prof. Qin Jin. My long-term goal is to build general-purpose humanoid robots that can perceive, reason, and act in the physical world. For more details about our recent research, please see the [BeingBeyond Research Blog](#).

EDUCATION

Renmin University of China

Ph.D. in Computer Science

2018.09 – 2023.06

Advisor: Prof. Qin Jin

Renmin University of China

B.S. in Computer Science

2014.09 – 2018.06

Advisor: Prof. Qin Jin

KEY PROJECTS

Being-H0.7: Latent World-Action Model (Project Lead; Co-first Author)

2026.04

Developed a latent world-action model that learns future-aware robot reasoning from 200,000 hours of egocentric human videos and 15,000 hours of robot demonstrations. The model uses learnable latent queries and a train-time posterior branch to absorb future interaction structure, then deploys as a direct action policy without pixel-space rollouts. It shows strong performance across simulation and real-robot tasks covering dynamic scenes, physical reasoning, motion reasoning, long-horizon control, and generalization.

Being-H0 & H0.5: VLA (Project Lead; Co-first Author)

2026.01

Built the Being-H VLA pretraining line from large-scale human videos to cross-embodiment robot control. Being-H0 learns dexterous manipulation via explicit hand-motion modeling and physical instruction tuning, while Being-H0.5 scales this direction with a unified action language and a Mixture-of-Transformers architecture. The system treats human hand motion as a first-class embodiment and transfers reusable manipulation structure across heterogeneous robots, control spaces, and real-world tasks.

Being-M0 & M0.5: Large Motion Models (Project Lead; Co-first Author)

2025.08

Developed real-time controllable vision-language-motion generation models trained on the HuMo100M data mixture. Being-M0.5 supports random instructions, initial-pose conditioning, long-term and unseen motion generation, and part-aware control, backed by over 5M self-collected motions and 100M multi-task instructional instances. A part-aware residual quantization design enables fine-grained control over individual body parts while keeping motion generation practical for interactive use.

Being-VL-0.5: Unified Vision-Language Model

2025.07

Built a unified multimodal understanding framework by applying Byte-Pair Encoding-style tokenization directly to visual tokens. The method merges frequent and spatially consistent VQ token pairs into structured visual units, then trains with a curriculum that progresses from visual-language alignment to perception, reasoning, and instruction following. This narrows the gap between discrete visual tokens and language tokens while preserving a unified token interface for multimodal foundation models.

Being-0: Agentic Embodied Framework

2025.02

Integrated a foundation model with a modular skill library: high-level instruction understanding, task planning, and reasoning are handled by the foundation model, while stable locomotion and dexterous manipulation are delegated to low-level skills.

WORK EXPERIENCE

BeingBeyond

2025.05 – Present

Partner, Research Scientist

Beijing, China

- Lead embodied multimodal pretraining across Being-H, Being-M, and Being-VL, covering VLA pretraining, human motion generation, multimodal modeling, and robot control.
- Scaled pretraining to billion-level image-text corpora and million-level video/motion datasets with self-built data curation pipelines.
- Deployed and evaluated models on humanoid platforms including Unitree G1, H1, H1-2, and Fourier GR1.

Beijing Academy of Artificial Intelligence

2023.07 – 2025.05

Research Scientist

Beijing, China

- Conducted research on multimodal agents, vision-language-audio pretraining, and embodied open-world control.
- Published work on embodied agents and multimodal representation learning at ICLR, NAACL, ECCV, and related venues.

Beijing Academy of Artificial Intelligence

2022.12 – 2023.06

Research Intern

Beijing, China

- Worked on multilingual language-vision-audio pretraining and large multimodal model research.

Microsoft Research Asia

2022.04 – 2022.10

Research Intern

Beijing, China

- Worked on temporal sentence grounding for long-form videos.

AWARDS

Ranked 1st, GemBench Challenge at CVPR 2025 GRAIL Workshop.	2025
Ranked 3rd, Facebook CVPR 2022 Ego4D Natural Language Query Challenge.	2022
Ranked 3rd, NIST TRECVID 2021 Ad-hoc Video Search Challenge.	2021
Ranked 2nd, CVPR 2021 HOMAGE Scene-Graph Generation Challenge.	2021
Ranked 2nd, ACM MM 2020 Video Relationship Understanding Grand Challenge.	2020
Ranked 2nd, ACM MM 2019 Video Relationship Understanding Grand Challenge.	2019
National Scholarship for Ph.D. Students, Top 5%.	2022
Best Method Prize, ACM MM 2019 Grand Challenge.	2019
First-Class Scholarship for Ph.D. Students, Top 10%.	2018–2021
First Prize, National University Mathematical Modeling Competition.	2015

PUBLICATIONS

† denotes equal contribution.

1. BeingBeyond Team. Being-H0.7: A Latent World-Action Model from Egocentric Videos. *arxiv*, 2026
2. Haiweng Xu, **Sipeng Zheng**, Hao Luo, Wanpeng Zhang, Ziheng Xi, and Zongqing Lu. Unmasking the Illusion of Embodied Reasoning in Vision-Language-Action Models. *arXiv*, 2026
3. Wanpeng Zhang, Hao Luo, **Sipeng Zheng**, Yicheng Feng, Haiweng Xu, Ziheng Xi, Chaoyi Xu, Haoqi Yuan, and Zongqing Lu. Conservative Offline Robot Policy Learning via Posterior-Transition Reweighting. *arXiv*, 2026
4. BeingBeyond Team. Being-H0.5: Scaling Human-Centric Robot Learning for Cross-Embodiment Generalization. *arxiv*, 2026
5. Wanpeng Zhang, Ye Wang, Hao Luo, Haoqi Yuan, Yicheng Feng, **Sipeng Zheng**, Qin Jin, and Zongqing Lu. DiG-Flow: Discrepancy-Guided Flow Matching for Robust VLA Models. *arxiv*, 2026
6. Junpeng Yue, Zepeng Wang, Yuxuan Wang, Weishuai Zeng, Jiangxing Wang, Xinrun Xu, Yu Zhang, **Sipeng Zheng**, Ziluo Ding, and Zongqing Lu. RL from Physical Feedback: Aligning Large Motion Model with Robot Whole Body Control. *arxiv*, 2026
7. Boyuan Li, **Sipeng Zheng**, Bin Cao, Ruihua Song, and Zongqing Lu. Robust Motion Generation using Part-level Reliable Data from Videos. *arxiv*, 2026
8. Bin Cao, **Sipeng Zheng**, Hao Luo, Boyuan Li, Jing Liu, and Zongqing Lu. OpenT2M: No-frill Motion Generation with Open-source, Large-scale, High-quality Data. *CVPR*, 2026
9. BeingBeyond Team. Being-H0: Vision-Language-Action Pretraining from Large-Scale Human Videos. *ICML*, 2026
10. Yicheng Feng, Wanpeng Zhang, Ye Wang, Hao Luo, Haoqi Yuan, **Sipeng Zheng**, and Zongqing Lu. Spatial-Aware VLA Pretraining through Visual-Physical Alignment from Human Videos. *CVPR*, 2026
11. Hao Luo, Ye Wang, Wanpeng Zhang, Haoqi Yuan, Yicheng Feng, Haiweng Xu, **Sipeng Zheng**, and Zongqing Lu. Predictive Embedding as Latent Action: Towards VLA Pretraining in the Wild. *CVPR*, 2026
12. Hao Luo, Zihao Yue, Wanpeng Zhang, Yicheng Feng, **Sipeng Zheng**, Deheng Ye, and Zongqing Lu. OpenMMEgo: Enhancing Egocentric Understanding for LMMs with Open Weights and Data. *NeurIPS*, 2025
13. Boshen Xu, Yuting Mei, Xinbi Liu, **Sipeng Zheng**, and Qin Jin. EgoDTM: Towards 3D-Aware Egocentric Video-Language Pretraining. *NeurIPS*, 2025
14. Jiazheng Liu, Börje F. Karlsson, **Sipeng Zheng**, and Zongqing Lu. Taking Notes Brings Focus? Towards Multi-Turn Multimodal Dialogue Learning. *EMNLP*, 2025
15. Yuting Mei, Ye Wang, **Sipeng Zheng**, and Qin Jin. Integrating Path Planning and Adaptive Locomotion for Mobile Quadruped Robots with Large Multimodal Models. *arxiv*, 2025
16. Wanpeng Zhang, Yicheng Feng, Hao Luo, Yijiang Li, Zihao Yue, **Sipeng Zheng**, and Zongqing Lu. Unified Multimodal Understanding via Byte-Pair Visual Encoding. *ICCV*, 2025 (Highlight)
17. Bin Cao[†], **Sipeng Zheng**[†], Ye Wang, Lujie Xia, Qianshan Wei, Qin Jin, Jing Liu, and Zongqing Lu. MotionCtrl: A Real-time Controllable Vision-Language-Motion Model. *ICCV*, 2025
18. Yicheng Feng, Yijiang Li, Wanpeng Zhang, **Sipeng Zheng**, and Zongqing Lu. VideoOrion: Tokenizing Object Dynamics in Videos. *ICCV*, 2025
19. Ye Wang[†], **Sipeng Zheng**[†], Bin Cao, Qianshan Wei, Weishuai Zeng, and Zongqing Lu. Being-M0: Scaling Large Motion Models with Million-Level Human Motions. *ICML*, 2025

20. Wanpeng Zhang, Zilong Xie, Yicheng Feng, Yijiang Li, Xingrun Xing, **Sipeng Zheng**, and Zongqing Lu. From Pixels to Tokens: Byte-Pair Encoding on Quantized Visual Modalities. *ICLR*, 2025
21. Boshen Xu, Ziheng Wang, Yang Du, Zhinan Song, **Sipeng Zheng**, and Qin Jin. EgoNCE++: Do Egocentric Video-Language Models Really Understand Hand-Object Interactions? *ICLR*, 2025
22. Boshen Xu, **Sipeng Zheng**, and Qin Jin. SPAFormer: Sequential 3D Part Assembly with Transformers. *3DV*, 2025
23. **Sipeng Zheng**, Bohan Zhou, Yicheng Feng, Ye Wang, and Zongqing Lu. UniCode: Learning a Unified Codebook for Multimodal Large Language Models. *ECCV*, 2024
24. **Sipeng Zheng**, Jiazheng Liu, Yicheng Feng, and Zongqing Lu. Steve-Eye: Equipped LLM-based Embodied Agents with Visual Perception in Open Worlds. *ICLR*, 2024 (**Spotlight 5.02%**)
25. Yicheng Feng, Yuxuan Wang, Jiazheng Liu, **Sipeng Zheng**, and Zongqing Lu. LLaMA Rider: Spurring Large Language Models to Explore the Open Worlds. *NAACL*, 2024
26. Qi Zhang, **Sipeng Zheng**, and Qin Jin. No-frills Temporal Video Grounding: Multi-Scale Neighboring Attention and Zoom-in Boundary Detection. In *AAAI*, 2023
27. **Sipeng Zheng**, Boshen Xu, and Qin Jin. Open-Category Human-Object Interaction Pre-training via Language Modeling Framework. In *CVPR*, 2023
28. Boshen Xu, **Sipeng Zheng**, and Qin Jin. POV: Prompt-Oriented View-agnostic Learning for Egocentric Hand-Object Interaction in the Multi-view World. *ACM MM*, 2023
29. Ludan Ruan, Anwen Hu, Yuqing Song, Liang Zhang, **Sipeng Zheng**, and Qin Jin. Accommodating audio modality in CLIP for multimodal processing. *AAAI*, 2023
30. **Sipeng Zheng**, Shizhe Chen, and Qin Jin. Few-shot Action Recognition with Hierarchical Matching and Contrastive Learning. In *ECCV*, 2022
31. **Sipeng Zheng**, Shizhe Chen, and Qin Jin. VRDFormer: End-to-End Video Visual Relation Detection With Transformers. In *CVPR*, 2022 (**Oral 4.14%**)
32. **Sipeng Zheng**, Qi Zhang, and Qin Jin. Exploring Anchor-based Detection for Ego4D Natural Language Query. In *CVPR Ego4D workshop*, 2022
33. Bei Liu, **Sipeng Zheng**, Jianlong Fu, and Wen-Huang Cheng. Anchor-Based Detection for Natural Language Localization in Ego-Centric Videos. In *IEEC*, 2022
34. **Sipeng Zheng**, Shizhe Chen, and Qin Jin. Skeleton-Based Interactive Graph Network For Human Object Interaction Detection. In *ICME*, 2020
35. **Sipeng Zheng**, Shizhe Chen, and Qin Jin. Visual Relation Detection with Multi-Level Attention. In *ACM MM*, 2019
36. **Sipeng Zheng**, Xiangyu Chen, Shizhe Chen, and Qin Jin. Relation Understanding in Videos. In *ACM MM*, 2019

PROFESSIONAL SERVICE

- Conference reviewer for CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, and related venues.
- Journal reviewer for IJCV, TCSVT, TMM, and JATS.